ASSIGNMENT 4:

SEVERITY AND CASE CLASSIFICATION

1. Introduction

Severity and case classification are important steps in pharmacovigilance. They help determine the clinical impact of adverse drug reactions (ADRs) and identify cases that may require urgent reporting.

ADR cases are assessed based on three main factors:

- Severity
- Seriousness
- Expectedness

2. Definitions

Severity: Refers to the intensity of an adverse reaction. It is generally classified as mild, moderate, or severe.

Seriousness: Refers to whether an adverse event meets specific regulatory criteria, such as death, life-threatening condition, hospitalisation, disability, congenital anomaly, or another medically important event.

Expectedness: Determines whether the nature and severity of the reaction are consistent with the known safety information of the suspected medicine.

3. Classification of ADRs

A. Severity Classification

Category	Description
Mild	Reaction causes minimal discomfort and usually requires no specific treatment.
Moderate	Reaction causes noticeable discomfort and may require medical treatment.
Severe	Reaction is intense, significantly affects daily activities, or requires substantial medical intervention.

B. Seriousness Classification

A reaction is considered serious if it results in or involves any of the following:

1. Death.
2. Life-threatening condition.
3. Initial or prolonged hospitalisation.
4. Persistent or significant disability or incapacity.
5. Congenital anomaly or birth defect.
6. Other medically important events requiring intervention to prevent serious outcomes.

C. Expectedness Classification

Category	Description
Expected	The reaction is consistent with the known safety information of the medicine.
Unexpected	The reaction is not consistent with the known safety information, or its nature or severity differs from what is documented.

4. Case Evaluation and Classification

The following are *hypothetical* cases created for this assignment.

Case 1: Mild Adverse Drug Reaction

Patient details: Adult patient taking cetirizine for allergic rhinitis.

Reaction: The patient experiences mild drowsiness after taking the medicine. No medical intervention or hospitalisation is required.

AssessmentClassification

Severity	Mild
Seriousness	Non-serious
Expectedness	Expected

Justification: Drowsiness is a known adverse effect of cetirizine. The reaction is mild and does not meet any seriousness criteria.

Case 2: Serious Adverse Drug Reaction

Patient details: Adult patient taking ibuprofen for pain relief.

Reaction: The patient develops gastrointestinal bleeding and requires hospitalisation and medical treatment.

AssessmentClassification

Severity	Severe
Seriousness	Serious
Expectedness	Expected

Justification: Gastrointestinal bleeding is a known risk associated with ibuprofen. The reaction is severe and requires hospitalisation, making it a serious adverse drug reaction.

Case 3: Unexpected Adverse Drug Reaction

Patient details: Patient participating in a clinical trial of a new investigational medicine.

Reaction: The patient develops persistent hearing loss after receiving the investigational medicine. The reaction is not listed in the available Investigator's Brochure safety information.

AssessmentClassification

Severity	Severe
Seriousness	Serious
Expectedness	Unexpected

Justification: Persistent hearing loss results in significant disability. Since the reaction is not documented in the available safety information, it is classified

as unexpected. The case meets the seriousness criterion of significant disability.

5. **Summary of Case Classification**

Case	Suspected Reaction	Severity	Seriousness	Expectedness
1	Drowsiness with cetirizine	Mild	Non-serious	Expected
2	Gastrointestinal bleeding with ibuprofen	Severe	Serious	Expected
3	Persistent hearing loss with investigational medicine	Severe	Serious	Unexpected

6. **Conclusion**

Severity, seriousness, and expectedness are essential factors in pharmacovigilance case classification. Proper assessment helps identify clinically important adverse reactions, supports patient safety monitoring, and assists in determining appropriate reporting requirements.

7. **References**

- International Council for Harmonisation (ICH). E2A: Clinical Safety Data Management – Definitions and Standards for Expedited Reporting. 1995.
- U.S. FDA. What is a Serious Adverse Event?